Evaluate Yourself

6) 0.16 g of an organic compound was heated in a carius tube and H_2SO_4 acid formed was precipitated with $BaCl_2$. The mass of $BaSO_4$ was 0.35g. Find the percentage of sulphur

Estimation of halogens: carius method: A known mass of the organic compound is heated with fuming HNO_3 and $AgNO_3$. C,H &S get oxidized to CO_2, H_2O & SO_2 and halogen combines with $AgNO_3$ to form a precipitate of silver halide.

$$X \xrightarrow[AgNO_3]{fum.HNO_3} AgX\downarrow.$$

The ppt of AgX is filtered,

washed, dried and weighed. From the mass of AgX and the mass of the organic compound taken, percentage of halogens are calculated.

A known mass of the substance is taken along with fuming HNO_3 and $AgNO_3$ is taken in a clean carius tube. The open end of the Carius tube is sealed and placed in a iron tube for 5 hours in the range at 530-540 K Then the tube is allowed to cool and a small hole is made in the tube to allow gases produced to escape. The tube is broken and the ppt is filtered, washed, dried and weighed. From the mass of AgX obtained, percentage of halogen in the organic compound is calculated.

Weight of the organic compound: w g

Weight of AgCl precipitate = a g

143.5 g of AgCl contains 35.5 g of Cl

$\therefore$ a g of AgCl contains $\frac{35.5}{143.5} \times a$

w g Organic compound gives a g AgCl

Percentage of Cl in w g $= \left(\frac{35.5}{143.5} \times \frac{a}{w} \times 100\right)\%$ organic compound

Let Weight of silver Bromide be 'b'g

188g of AgBr contains 80 g of Br

$\therefore$ b g of AgBr contains $\left(\frac{80}{188}\ b\right)$ g of Br

w g Organic compound gives b g AgBr

Percentage of Br in w g $= \left(\frac{80}{188} \times \frac{b}{w} \times 100\right)\%$ organic compound

Let Weight of silver Iodide be 'c'g

235g of AgI contains 127 g of I

$\therefore$ C g of AgI contains $\left(\frac{127}{235}\ c\right)$ g of I

w g Organic compound gives c g AgI

Percentage of I in w g $= \left(\frac{127}{235} \times \frac{c}{w} \times 100\right)\%$ organic compound

EXAMPLE : 0.284 g of an organic substance gave 0.287 g AgCl in a carius method for the estimation of halogen. Find the Percentage of Cl in the compound.

Weight of the organic substance = 0.284 g

Weight of AgCl is = 0.287 g

143.5 g of AgCl contains 35.5 g of chlorine

0.287 g of AgCl contains $\frac{35.5}{143.5} \times \frac{0.287}{0.284}$

% of chlorine is $\frac{35.5}{143.5} \times \frac{0.287}{0.284} \times 100 = 24.98\%$

Unit 11 revised - jagan.indd 143 22-01-2021 12:42:29